

sLITERATURE REVIEW

StockSense

AI-Powered Inventory Shortage Predictor for Small Businesses

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Project Guide

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1. Introduction

This literature review surveys existing research and industry approaches related to inventory management, demand forecasting, and machine-learning-based shortage prediction. The goal is to understand current techniques, identify their strengths and limitations in the context of small businesses, and position StockSense relative to existing work.

2. Reviewed Works

2.1 Traditional Inventory Management Systems

Conventional inventory management systems focus on record-keeping — tracking stock-in, stock-out, and reorder levels — rather than forecasting. These systems typically use static reorder-point rules (e.g., reorder when stock falls below a fixed number) rather than adaptive, data-driven predictions. While reliable for simple use cases, static rules do not account for seasonal demand shifts, promotional spikes, or changing customer behavior, which limits their usefulness for businesses with variable demand.

2.2 Statistical Demand Forecasting Methods

Classical forecasting techniques such as Moving Average, Exponential Smoothing, and ARIMA (AutoRegressive Integrated Moving Average) have long been used to forecast product demand from historical sales time-series. These methods perform reasonably well for stable, trend-driven demand patterns but tend to struggle with sparse data, irregular purchasing patterns, and multiple interacting factors — conditions that are common for small businesses with limited transaction history.

2.3 Machine Learning-Based Demand and Stockout Prediction

More recent research applies supervised machine learning models — such as Random Forest, Gradient Boosting, and regression-based approaches — to demand and stockout prediction. These models can incorporate multiple features (day of week, seasonality, promotions, historical sales velocity) and generally outperform purely statistical methods when sufficient historical data is available. However, many published approaches are evaluated on large retail or e-commerce datasets, and their applicability to small businesses with limited data volume and no dedicated data science resources is less explored.

2.4 Deep Learning Approaches (LSTM/RNN-based Forecasting)

Deep learning models, particularly Long Short-Term Memory (LSTM) networks, have been explored for sequential sales forecasting due to their ability to capture long-term temporal dependencies. While these models can achieve strong accuracy on large, data-rich retail chains, they require substantial training data and computational resources — making them impractical for a lightweight, small-business-focused tool such as StockSense.

2.5 Inventory Management Mobile & Web Applications (Industry Tools)

Several commercial inventory tools (e.g., point-of-sale add-ons and standalone inventory apps) offer stock tracking and low-stock notifications. Most rely on simple threshold-based alerts rather than predictive modeling, and many are priced or featured for medium-to-large retailers, leaving a gap for an affordable, prediction-driven tool tailored to small businesses.

2.6 AI Adoption Challenges in Small Businesses

Studies on technology adoption among small and medium enterprises (SMEs) highlight barriers such as limited technical expertise, cost sensitivity, and lack of clean historical data. This underscores the need for prediction systems that are not only accurate but also simple to use, affordable, and tolerant of imperfect data — a key design consideration for StockSense.

3. Comparative Summary

Approach	Strengths	Limitations
Traditional / Rule-Based Systems	Simple, easy to implement, low resource requirement	No adaptability to changing demand patterns
Statistical Methods (ARIMA, Moving Average)	Well-established, interpretable, works with moderate data	Struggles with sparse/irregular data, ignores multiple influencing factors
Classical Machine Learning (Random Forest, Gradient Boosting)	Handles multiple features, good accuracy with moderate data	Requires careful feature engineering, moderate data requirement
Deep Learning (LSTM/RNN)	Captures complex temporal patterns, high accuracy at scale	Needs large datasets & compute, hard to deploy for small businesses
Commercial Inventory Apps	Ready to use, user-friendly interfaces	Mostly threshold-based, not predictive; often costly for small businesses

4. Research Gap

Across the reviewed literature, two consistent gaps emerge for small-business inventory management:

- Most predictive (ML/deep learning) approaches are designed and evaluated for large retailers with abundant data, making them difficult to apply directly to small businesses with limited historical records.
- Most tools accessible to small businesses rely on static, threshold-based alerts rather than genuine predictive forecasting, leaving owners to react only after a shortage is imminent or has already occurred.

StockSense aims to address this gap by using lightweight, data-efficient machine learning models suited to smaller datasets, combined with a simple, affordable web dashboard — bringing predictive shortage forecasting within reach of small business owners who currently rely on manual tracking or basic threshold alerts.

5. References

Note: The following references represent illustrative source types typically cited in this domain. The final document will be updated with the specific papers, articles, and sources consulted by the team, formatted in the citation style prescribed by the department.

- Research literature on statistical time-series forecasting methods (e.g., ARIMA, Exponential Smoothing) for demand prediction.
- Research literature on machine learning approaches (Random Forest, Gradient Boosting, Regression) applied to retail demand and stockout prediction.
- Research literature on deep learning (LSTM/RNN) approaches for sequential sales forecasting.

- Industry reports and case studies on inventory management tools for small and medium enterprises.
- Studies on technology and AI adoption challenges among small and medium businesses.